

Collective Intelligence — Module 04: Cognition —

Discussion Questions

20 unique questions on shared mental models, group sensemaking, cognitive diversity, and groupthink.

1. What is the difference between "shared mental models" and "groupthink"? How do you maintain the benefits of alignment while preserving diversity?
2. Weick's analysis of the Mann Gulch disaster shows that when shared mental models collapse, teams disintegrate. What causes a shared model to collapse? What can prevent it?
3. How does cognitive diversity contribute to better team problem-solving? When does diversity become a barrier to team function?
4. Design a "pre-mortem" exercise for a team about to launch a major project. How does imagining failure before it happens improve the group's generative model?
5. How do after-action reviews (AARs) enable collective model updating? What conditions make AARs effective versus performative?
6. Compare consensus-building with majority rule as decision methods. Which better supports shared cognition? Which better preserves cognitive diversity?
7. How does team tenure (how long members have worked together) affect shared mental models? What are the risks of very long-tenured teams?
8. Explain how "structured debate" can improve team cognition. Design a debate format for evaluating two competing strategic options.
9. A team member has a strong dissenting view but is reluctant to share it. What team conditions would make sharing more likely? How does psychological safety relate to collective cognition?

10. How do visual artifacts (whiteboards, diagrams, shared documents) support shared cognition? What cognitive function do they serve?
11. Map your team's shared mental model. What beliefs do all members hold in common? Where do individual models diverge? Are the divergences productive or destructive?
12. Identify a recent team decision that was influenced by groupthink. What were the warning signs? What would you change about the process to prevent it?
13. Design a cognitive diversity assessment for your team — not demographic diversity, but diversity of mental models, expertise, and problem-solving approaches.
14. Your team made a decision that turned out wrong. Conduct a "cognitive post-mortem": What was the team's shared model at the time? What evidence was available but not incorporated? What would have changed the outcome?
15. How should a team handle a situation where the data supports one conclusion but the most experienced member's intuition says otherwise?
16. Compare how a newly formed team and an established team approach sensemaking for an ambiguous situation. What are the advantages of each?
17. How do cultural differences affect shared mental model formation? What challenges arise in multicultural teams?
18. Design an onboarding process that explicitly builds a new member into the team's shared mental model rather than just teaching task skills.
19. When is it appropriate to deliberately introduce cognitive conflict into a team? How would you manage it constructively?
20. A team has been working together so long that they finish each other's sentences. Is this a sign of excellent shared cognition or dangerous groupthink? How would you tell the difference?